TECHNICAL DOCUMENTATION

HR Employee Attrition & Retention Dashboard

A Power BI analysis of why employees leave and who is most at risk

TEAM

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Tools: Power Query · DAX · Power BI Desktop | Version 1.0 – October 2026

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Items marked **TODO** need input from the team member who owns that part. Items marked *illustrative* are proposed from what the dashboard shows and must be checked against the .pbix.

1. Purpose & Scope

This document explains how the HR Employee Attrition & Retention Dashboard was built technically: data preparation, model, measures, KPIs, report structure, and the testing done before delivery. The business context, findings and recommendations are in *Project_Booklet.pdf*.

2. Tools Used per Stage

Stage	Tool	Output
Data cleaning & preparation	Power Query (Power BI)	HR_Employee_Attrition_Cleaned
Data modeling	Power BI Desktop (Model view)	Fact_Employee, Dim_Job, Measures_HR, Progress Axis
Measures & KPIs	DAX	Measures_HR table
Dashboard design	Power BI Desktop	7-page .pbix report
Documentation & presentation	PDF, PowerPoint	Booklet, technical doc, slides
Version control	Git & GitHub	Organized repository

3. Data Cleaning & Power Query

Source: IBM HR Analytics Employee Attrition & Performance (Kaggle, fictional data). Source file: *HR_Employee_Attrition_Cleaned* (1,470 rows, **32 columns**, one row per employee). Checked on the file: 0 missing values, 0 duplicate rows, 24 numeric and 8 text columns, all names in snake_case.

Preparation steps reconstructed from the cleaned file (to be confirmed by Sandra, Person 1):

#	Step	Evidence
1	Remove constant columns EmployeeCount, Over18, StandardHours	Original dataset has 35 columns; cleaned file has 32
2	Rename columns to snake_case	e.g. job_level, monthly_income, over_time
3	Clean text values	business_travel uses spaces: Travel Rarely, Travel Frequently, Non-Travel
4	Check quality	0 missing values, 0 duplicate rows, employee_number unique
5	Create analysis bands	Age group, income band, tenure group, years-since-promotion (not in the 32 columns, so created in Power BI)

TODO – team input needed: Sandra (Person 1): confirm the steps above and send a screenshot of Power Query → Applied Steps.

Evidence of preparation visible in the report

- Attrition stored as a readable Left / Stayed outcome.
- Column names such as *job_level* appear in snake_case in the Tenure table, which suggests columns were renamed in Power Query.
- Categorical codes labeled: Job Satisfaction 1 – Low to 4 – Very High; business travel as Travel Frequently / Travel Rarely / Non-Travel.

4. Data Model: Tables & Relationships

Table	Type	Description
Fact_Employee	Fact	One row per employee: demographics, income, tenure, promotion, overtime, travel, satisfaction, stock option, attrition status
Dim_Job	Dimension	Department, job role, job level
Measures_HR	Measures only	Holds all DAX measures, no data rows
Progress Axis	Helper	Supports dashboard visuals (e.g. progress/axis bars)

Design notes

- Star-style model: Fact_Employee in the center, Dim_Job as the lookup for job attributes.
- All measures are kept in one table so they are easy to find and reuse.

TODO – team input needed: Ahmed (Person 2): relationships table (From column → To column, cardinality, filter direction, active/inactive) and the Data Model screenshot from Model view.

5. Derived Columns (Analysis Bands)

These groupings are used by the charts. Their boundaries are taken from the dashboard labels.

Column	Bands
Age group	Under 30 · 30-39 · 40-49 · 50+
Income band	Under 3K · 3K-6K · 6K-10K · 10K+
Tenure group	0-1 · 2-5 · 6-10 · 10+ years
Years since last promotion	0 · 1 · 2 · 3 · 4-5 · 6+

Verified against the dataset: years at company are whole numbers, so the bands are 0-1, 2-5, 6-10 and more than 10 (counts: 215, 561, 448, 246). There is no gap and no overlap, but the last label “10+” is misleading: employees with exactly 10 years sit in 6-10. Rename it “11+”.

6. DAX Measures

Proposed definitions, *illustrative*. Table and column names are placeholders and must be replaced with the real ones from **Measures_HR**.

```
Total Employees = COUNTROWS ( Fact_Employee )
Employees Left = CALCULATE ( [Total Employees], Fact_Employee[Attrition] = "Yes" )
Active Employees = [Total Employees] - [Employees Left]
Attrition Rate = DIVIDE ( [Employees Left], [Total Employees] )
Retention Rate = DIVIDE ( [Active Employees], [Total Employees] )
Company Attrition Rate = CALCULATE ( [Attrition Rate], ALL ( Fact_Employee ) )
Avg Monthly Income = AVERAGE ( Fact_Employee[MonthlyIncome] )
Avg Income - Attrited = CALCULATE ( [Avg Monthly Income], Fact_Employee[Attrition] = "Yes" )
Avg Income - Active = CALCULATE ( [Avg Monthly Income], Fact_Employee[Attrition] = "No" )
Avg Years at Company = AVERAGE ( Fact_Employee[YearsAtCompany] )
Avg Years Since Promotion = AVERAGE ( Fact_Employee[YearsSinceLastPromotion] )
Avg Age = AVERAGE ( Fact_Employee[Age] )
Overtime Attrition Rate = CALCULATE ( [Attrition Rate], Fact_Employee[OverTime] = "Yes" )
```

Measure	What it does	Used on
Attrition Rate	Core KPI. Reused by every “Attrition Rate by …” chart; the chart’s category supplies the filter context.	All pages
Company Attrition Rate	Ignores filters to give the 16.12% reference line.	Overview (department chart)
Avg Income - Active / Attrited	Compares pay of stayers and leavers.	Compensation
Overtime Attrition Rate	Attrition among overtime workers only.	Drivers

TODO – team input needed: Ahmed (Person 2): paste the real DAX of each measure from the .pbix and note any extra measures (conditional-format colors, Progress Axis, insight-card text).

7. KPI Definitions & Calculation

KPI	Formula	Value	Check
Total Employees	Count of employees	1,470	–
Employees Left	Count where Attrition = Yes	237	–
Active Employees	Total – Left	1,233	1,470 – 237
Attrition Rate	Left ÷ Total	16.12%	237 ÷ 1,470 = 16.12%
Retention Rate	Active ÷ Total	83.9%	1,233 ÷ 1,470 = 83.88%
Avg Monthly Income	Average income	$6,503	–
Avg Income – Attrited	Average income of leavers	$4,787	–
Avg Years at Company	Average tenure	7.01	–
Avg Years Since Promotion	Average	2.2	–
Avg Age	Average	36.9	–
Overtime Attrition Rate	Left ÷ Total among overtime = Yes	30.5%	127 ÷ 416 = 30.53%

Rule: attrition rate is always **employees who left ÷ total employees** for the group shown. Groups under 100 employees (HR, Sales Representative, single + overtime) should be read with caution.

8. Report Structure & Visuals

Page	KPI cards	Visuals
Overview	Total Employee, Active Employees, Attrition Rate, Avg Per Employee, Avg Age Years	Attrition by Job Level (bar); by Department with company-rate line (column); by Age Group (column); Left vs Stayed (donut); Department table (employees, attrition rate, avg income)
Drivers	Total Employees, Attrition Rate, Active Employees, Overtime Attrition	Attrition by Tenure, Income Band, Business Travel, Overtime (columns); by Marital Status (bar)
Compensation	Average Monthly Income, Attrition Rate, Avg Years at Company, Avg Income – Attrited	Attrition by Job Role (bar); Avg Income by Job Level, active vs attrited (clustered column); Avg Income by Department; Attrition by Stock Option level and Job Satisfaction
Tenure	Total Employees, Attrition Rate, Avg Years at Company, Avg Years Since Promotion	Avg Years by Job Level; Attrition by Years Since Last Promotion; Headcount by Tenure Group split by gender (stacked); Attrition by Companies Worked; job-level table
Workforce	Total, Active, Retention Rate, Attrition Rate	Headcount by Job Role split by department; by Age Group; by Education Field; Marital Status; Gender donut
Insights	Total Employees, Employees Who Left, Company Attrition Rate, Highest-Risk Group	Eight insight cards, each with a figure, explanation and recommended action; footer with source and definition

Color logic: the highest value in a chart is highlighted in pink, with purple for the rest, and mint/gold for the lowest in some charts. The palette follows the dashboard theme (dark green, mint, pink, lavender, gold).

9. Navigation & Interactivity

- Left-hand navigation buttons (1.Overview to 5.Workforce, plus 6.Insights on the Insights page) move between pages.
- Each page has a filter icon and a clear-filter icon in the header.
- Charts cross-filter each other, so selecting a bar filters the KPI cards and other visuals on the page.
- The Home page has buttons to jump to Overview, Drivers, Compensation, Tenure and Workforce.

TODO – team input needed: Haneen (Person 3): list the slicers actually used (e.g. department, gender, job level), the bookmarks behind the filter buttons, and edit-interactions settings.

10. Testing & Validation

Checks done by recomputing the dashboard numbers from the counts shown on it:

Figure	Recomputed	Result
Company attrition	237 ÷ 1,470 = 16.12%	Matches
0-1 years tenure	75 ÷ 215 = 34.9%	Matches
Overtime	127 ÷ 416 = 30.5%	Matches
Sales Representative	33 ÷ 83 = 39.8%	Matches
Sales department	92 ÷ 446 = 20.6%	Matches
Income under $3K	113 ÷ 395 = 28.6%	Matches
Under 30	91 ÷ 326 = 27.9%	Matches
Frequent travelers	69 ÷ 277 = 24.9%	Matches
Single employees	120 ÷ 470 = 25.5%	Matches
Job level counts	543 + 534 + 218 + 106 + 69 = 1,470	Matches
Stayed vs left	83.9% + 16.1% = 100%	Matches

11. Known Issues & Final Improvements

Found while reviewing the dashboard. Each should be fixed in the .pbix by the dashboard owner, then this list updated.

#	Issue	Fix
1	Insights footer says “left + total employees”	Change to “left ÷ total employees”
2	Tenure label “10+” really means more than 10 years (10 exactly is in 6-10)	Rename the label to “11+” or “Over 10”
3	Donut centers show “1K” (Left vs Stayed, Gender Split)	Show the full number, 1,470
4	Avg Years at Company shows 7 on Compensation and 7.01 on Tenure	Use the same decimals
5	Same KPI named “Avg Per Employee” and “Average Monthly Income”	Use one name
6	Typos: “Jop Level”, “Ave Income”, “Attrition Rate by Companies Worked”	Correct the titles
7	“6.Insights” button missing on the other pages	Add it to every page
8	Stock option level 3 (17.6%) breaks the “more options, less attrition” pattern	Qualify the insight text or check the data

Appendix A. Data Dictionary

The 32 columns of HR_Employee_Attrition_Cleaned. Scales follow the public dataset documentation.

Column	Meaning
age	Age in years (18-60)
attrition	Employee left the company (Yes/No)
business_travel	Non-Travel, Travel Rarely, Travel Frequently
daily_rate	Daily pay rate (not used in the dashboard)
department	Sales, Research & Development, Human Resources
distance_from_home	Distance from home to work (1-29)
education	Level 1-5 (1 Below College to 5 Doctor)
education_field	Field of study (6 values)
employee_number	Unique employee ID
environment_satisfaction	1 Low to 4 Very High
gender	Female / Male
hourly_rate	Hourly pay rate (not used)
job_involvement	1 Low to 4 Very High
job_level	Level 1-5
job_role	Nine roles, e.g. Sales Representative
job_satisfaction	1 Low to 4 Very High
marital_status	Single, Married, Divorced
monthly_income	Monthly income in dollars (1,009-19,999)
monthly_rate	Monthly rate (not used)
num_companies_worked	Companies worked before (0-9)
over_time	Works overtime (Yes/No)
percent_salary_hike	Last salary increase in %
performance_rating	3 or 4 in this data
relationship_satisfaction	1 Low to 4 Very High
stock_option_level	0-3
total_working_years	Total career years
training_times_last_year	Trainings last year
work_life_balance	1 Bad to 4 Best
years_at_company	Tenure in years
years_in_current_role	Years in current role
years_since_last_promotion	Years since last promotion (0-15)
years_with_curr_manager	Years with current manager